Kigali Independent University (ULK)
Data Science

REPORT 1

Project Description Report

Nyalatna AI-Powered Training Center Management System

Nyalatna Capacity Building Center, Nyala, Sudan

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Assignment: UML Class Diagram Modelling Assignment

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Date: 19 October 2026

Repository: https://github.com/mbarkdlwd-blip/uml-class-diagram-assignment

1. Project Title and Business Idea

Project title: Nyalatna AI-Powered Training Center Management System

Nyalatna Capacity Building Center is a training center in Nyala, South Darfur, Sudan. It runs professional courses in numbered cohorts called batches and has already graduated 16 batches. The business idea is an information system that manages the full path of an applicant: asking about a course, registering in a batch, paying fees, having each payment checked by a second staff member, and receiving a certificate. An AI assistant answers applicant enquiries and collects registration details, so staff spend less time on repeated questions.

2. Problem Statement

The center currently manages most of its work manually. This creates the following problems:

- Applicant details, course schedules, registrations and payments are kept in separate places (paper forms, spreadsheets and messaging apps), which leads to duplicated or incomplete records.
- Staff cannot easily see whether a batch still has free seats, so batches can be overbooked.
- The status of a registration (pending, confirmed, completed) is often unclear to both staff and applicants.
- Payments, including instalments, are recorded by hand. There is no clear rule for who checks a payment, so one person can record and approve the same payment, which creates a risk of errors or fraud.
- Staff spend a lot of time answering the same questions from applicants about courses, fees and dates.
- Certificates are issued manually, without a reliable link to completion and payment status.

3. Purpose of the Proposed System

The purpose of the system is to centralise applicant, course, batch, registration, payment and certificate records in one model, enforce batch capacity, and protect financial records through separation of duties: the staff user who records a payment cannot be the one who verifies it. An AI assistant handles routine applicant enquiries but is deliberately kept away from any payment decision.

4. Project Scope

4.1 Functions covered (in scope)

- Applicant and staff user profiles, with shared contact details modelled through an abstract Person class.
- Staff user accounts with roles (admin, registration admin, courses and reports admin, registration officer).
- Course catalogue with description and fee.
- Training batches with dates, capacity and status, including checking available seats.
- Registration of applicants in batches and tracking of registration status and balance.
- Recording of payments (full or instalments) by an authorised staff user.

- Independent verification of each payment by a different authorised staff user, with decision and notes.
- Issuing and revoking certificates for completed, eligible registrations.
- An AI assistant that answers enquiries and collects applicant details to start a registration.

4.2 Functions excluded (out of scope)

- Building a working application or database: this project is limited to analysis and UML design.
- Attendance tracking and session scheduling.
- Full accounting, payroll, and integration with banks or online payment gateways.
- Hosting learning content (videos, quizzes) for online courses.
- Rental of the center's halls and facilities.

5. Main Classes and Their Responsibilities

The system is modelled with 10 classes and 6 enumerations. Their responsibilities are:

Class	Responsibility
Person *(abstract)*	Abstract parent holding the name and contact details shared by applicants and staff users.
Applicant	A person applying for or attending a course. Registers in batches and follows their status.
User	A staff account with a role. Its role decides whether it may record or verify payments.
Course	A training programme in the catalogue, with its description and fee.
TrainingBatch	One scheduled offering (cohort) of a course, with dates, capacity and status.
Registration	Links one applicant to one training batch and tracks the registration status and balance.
Payment	A payment (full or instalment) recorded by a staff user against a registration.
PaymentVerification	The independent review of a payment by a second staff user, with the decision and notes.
Certificate	Proof of completion issued for an eligible, completed registration.
AIAssistant	Answers applicant enquiries and collects registration details. It has no access to payment verification.
BatchStatus *(enum)*	Allowed values: PLANNED, OPEN, FULL, CLOSED
RegistrationStatus *(enum)*	Allowed values: PENDING, CONFIRMED, COMPLETED, CANCELLED
PaymentStatus *(enum)*	Allowed values: RECORDED, PENDING_VERIFICATION, VERIFIED, REJECTED
VerificationDecision *(enum)*	Allowed values: APPROVED, REJECTED
UserRole *(enum)*	Allowed values: ADMIN, REGISTRATION_ADMIN, COURSES_REPORTS_ADMIN, REGISTRATION_OFFICER
CertificateStatus *(enum)*	Allowed values: ISSUED, REVOKED

6. System Requirements

6.1 Functional requirements

ID	Requirement
FR1	The system shall create and update applicant profiles and prevent duplicate applicants.
FR2	The system shall manage staff user accounts with one role each and allow accounts to be deactivated.
FR3	The system shall maintain a course catalogue with description and fee.
FR4	The system shall create training batches for a course and refuse a registration when the batch is full.
FR5	The system shall register an applicant in a batch and track the registration status.
FR6	The system shall let an authorised user record a payment and mark it as pending verification.
FR7	The system shall let a different authorised user approve or reject the payment, with notes.
FR8	The system shall reject a verification when the verifier is the same user who recorded the payment.
FR9	The system shall issue a certificate only when the registration is completed and fully paid.
FR10	The AI assistant shall answer applicant enquiries and start registration conversations, and shall never verify payments.

6.2 Non-functional requirements

Quality	Requirement
Security	Role-based access control; only authorised roles may record or verify payments.
Auditability	Every payment and verification keeps who did it and when, for later review.
Usability	Simple forms for reception staff; Arabic and English labels.
Performance	Works over slow or unstable internet connections common in Nyala.
Maintainability	Clear class structure so new roles, statuses or course types can be added easily.

7. Conclusion

The proposed system replaces scattered paper and spreadsheet records with one consistent model of the center's registration and payment activity, and builds separation of duties into the design. The detailed class design, including attributes, methods, relationships and multiplicities, is presented in Report 2: UML Class Diagram Report.